COMPLEX ENGINEERING PROBLEM

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Abstract

This report presents the results and analysis of a jar test conducted to determine the optimum alum dosage required for effective turbidity removal in water samples collected from a residential society. The primary objective of the study was to design a sedimentation-cum-coagulation tank tailored for the community's water treatment needs, adhering to the guidelines set by WASA (Water and Sanitation Authority). A series of tests were conducted to evaluate key water quality parameters, including pH, turbidity, total suspended solids (TSS), total dissolved solids (TDS), salinity, and chemical oxygen demand (COD). The jar test results revealed that an alum dosage of 2 mg/L achieved the highest turbidity removal (250 FTU), indicating it as the most effective and economical concentration. The findings were used to guide the design dimensions of the treatment tank, ensuring both compliance with local regulatory standards and scalability based on the jar test proportions. This report also considers the environmental implications of the selected coagulant, ensuring sustainability alongside functionality.

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1 Introduction

1.1 Background

Access to safe and clean drinking water is one of the most fundamental necessities for public health and well-being. As urban populations grow, the demand for effective water treatment solutions becomes increasingly critical. Faisal Hills, a rapidly developing residential society located near Taxila, Pakistan, is one such urban area experiencing population growth and infrastructural expansion. Spanning an area of approximately **52,977,107 square feet** (**4,921,734 square meters**), Faisal Hills is home to a diverse mix of residential, commercial, and recreational zones, as depicted in the satellite image acquired via Google Earth (Figure 1). With the expansion of this locality, ensuring the supply of potable water that meets national and international quality standards has emerged as a key concern for developers and residents alike.

1.2 Purpose

The primary purpose of this study is to design an efficient **sedimentation-cum-coagulation tank** for treating the raw water supply in Faisal Hills. This includes the assessment of water quality through laboratory testing, particularly the **Jar Test**, which helps determine the most effective dosage of coagulant (alum) for turbidity removal. By establishing the optimum dosage, the design can be tailored to both environmental safety and economic feasibility. The final aim is to ensure that the treated water meets the regulatory standards set by the **Water and Sanitation Authority (WASA)**.

1.3 Scope

This report encompasses the entire procedure from sample collection to final tank design. Water samples were collected directly from Faisal Hills and subjected to various quality analyses, including measurements of pH, turbidity, total suspended solids (TSS), total dissolved

solids (TDS), conductivity, salinity, and chemical oxygen demand (COD). A laboratory-scale Jar Test was conducted using varying concentrations of alum to determine the optimum dosage for coagulation. The results from this test were used to proportionally design a full-scale sedimentation-cum-coagulation tank, ensuring compliance with WASA design standards. Additionally, the environmental impacts of the chosen coagulant were analyzed to assess long-term sustainability.

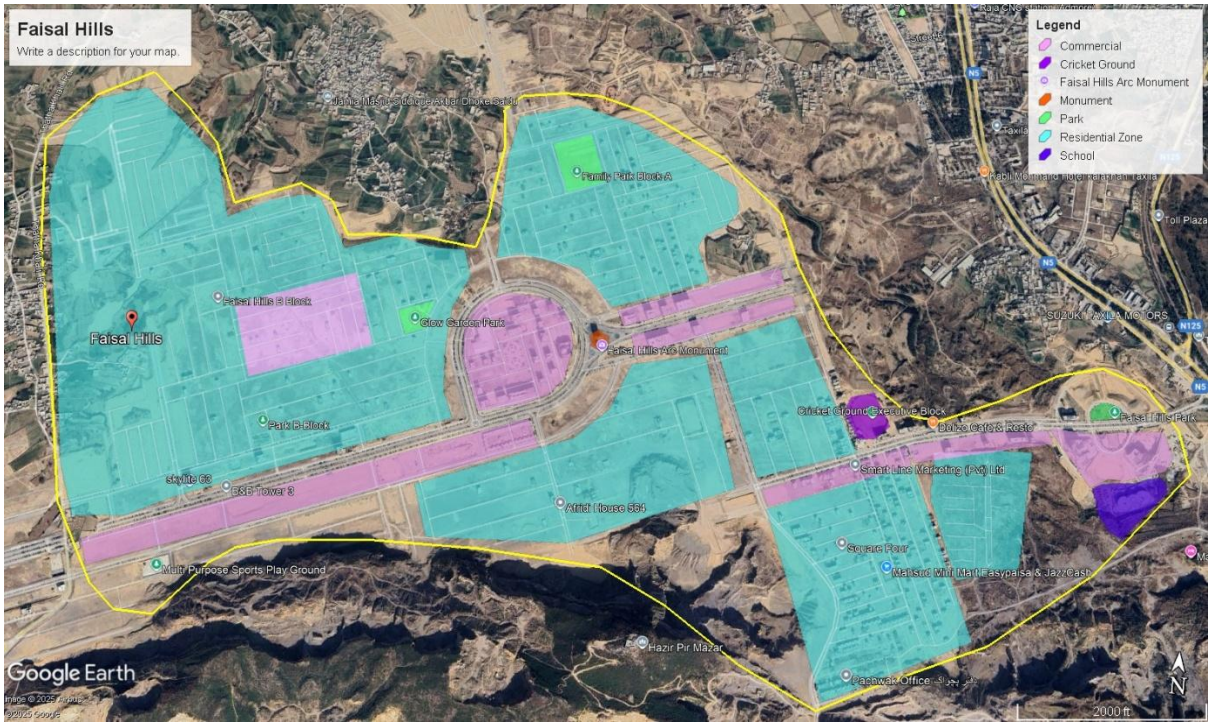


Figure 1 - Faisal Hills

2. Water Quality Analysis

2.1 pH Measurement

The pH of water indicates its acidity or alkalinity, which is crucial for evaluating its suitability for drinking and treatment processes. In this study, a calibrated digital pH meter was used to measure the pH of the water sample, ensuring accuracy and consistency in readings.

2.2 Turbidity

Turbidity measures the cloudiness of water caused by suspended particles. A nephelometric turbidity unit (NTU) meter, which detects light scattered by particles, was used to assess the turbidity before and after treatment to evaluate the effectiveness of the coagulant.

2.3 Total Dissolved Solids (TDS)

TDS indicates the concentration of dissolved substances such as minerals and salts in the water. A digital TDS meter was used to measure the value in mg/L, which helps in assessing water salinity and quality.

2.4 Total Suspended Solids (TSS)

TSS includes all particles suspended in water that are not dissolved. The sample was filtered, and the retained material was dried and weighed to determine the concentration of suspended solids in mg/L.

2.6 Salinity

Salinity refers to the total concentration of salts in water. It was measured using a salinity meter, providing results in mg/L as NaCl. This helps assess the water's compatibility with human consumption and agricultural use.

2.7 Chemical Oxygen Demand (COD)

COD indicates the amount of oxygen required to chemically oxidize organic pollutants in water. It was determined through a digestion process using a COD reactor and colorimetric analysis, providing insight into the organic contamination level of the water.

Table 1 Results of Water Quality Analysis and Comparison with Allowable Limits

Parameters	Units	Results	Maximum Allowable Limits
pH	pH units	7.23	**6.5-8.5**
Total dissolved solids	mg/L	413	**500mg/L**
Total suspended solids	mg/L	2	**10mg/L**
Turbidity	NTU	7	**5 NTU**
Salinity	mg/L	1000	**1000mg/l as NaCl**
Chemical oxygen demand	mg/L	6	**10mg/L**

3. Jar Test

3.1 Test Setup

The jar test was conducted using a standard jar test apparatus equipped with six beakers and mechanical stirrers. Equal volumes of the water sample were poured into each jar, and varying dosages of alum (aluminum sulfate, $Al_2(SO_4)_3$) were added. The water samples were stirred rapidly for a short period to promote dispersion of the coagulant, followed by slow mixing to facilitate flocculation. This process allowed for the observation and comparison of turbidity removal at different alum concentrations, leading to the determination of the optimum dosage. The test concluded that **2 mg/L of alum** is the most effective dosage for this particular water sample.

3.2 Results

Sample	Turbidity(FTU)	Turbidity(FTU)	Difference	Dosage	Difference
	Before	After			
1	273	23	250	2	250
2	214	53	161	4	161
3	195	18	177	6	177
4	177	43	134	8	134
5	169	74	95	10	95
6	183	26	157	12	157

3.3 Optimum Dosage

The graph illustrates (Table 1) the relationship between alum dosage and turbidity removal efficiency for the given water sample. It is evident that turbidity decreases significantly by increasing alum dosage up to a certain point. The optimum alum dosage for this water sample is identified as **2 mg/L**, where the maximum reduction in turbidity is achieved. Beyond this point, further increases in alum dosage do not result in substantial improvements in turbidity removal, indicating that 2 mg/L is the most efficient and economical dose for coagulation. in this case.

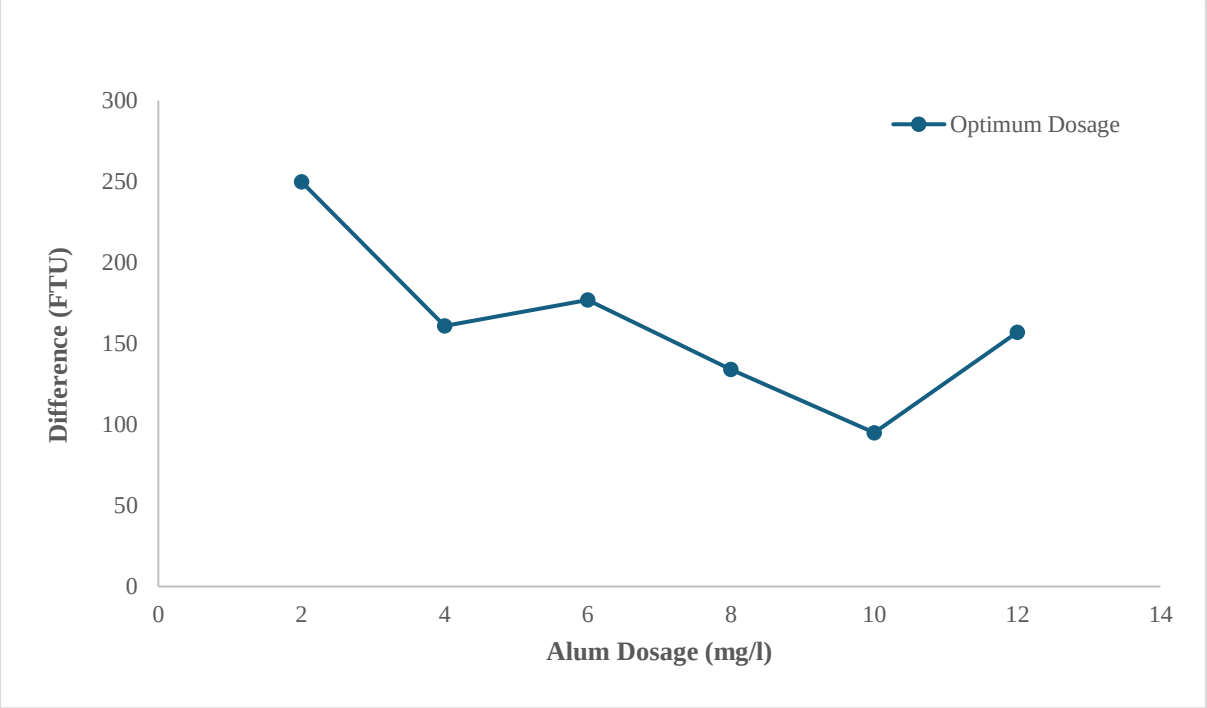


Table 1 - Optimum Dosage Content

4 Manual Calculations

5. Calculation and Recommendations

In this study, a water treatment system was analyzed for a population of approximately 150,740 people. The average daily water demand was calculated to be around 102,503,200 liters per day, with a maximum daily demand of 153,754,800 liters, equivalent to about 6,410 cubic meters per hour. While attempting to design the sedimentation tank, it was found that the required dimensions fell outside the permissible limits, rendering its design infeasible under the given constraints. However, a coagulation tank was successfully designed to treat the impurities in the water sample. The dimensions of the coagulation tank were 20 meters in breadth, 2 meters in depth, and approximately 54 meters in length. The optimum alum dosage was determined to be 2 mg/L based on turbidity removal efficiency. Environmentally, this dosage translates to an alum usage of approximately 30.75 kg, which is expected to result in the release of around 12 kg of CO_2 and contribute approximately 18.84 kg of $CaSO_4$ in terms of permanent hardness.

Re-evaluation of Sedimentation Tank Design. Since the sedimentation tank dimensions exceeded design constraints, it is recommended to reassess either the design parameters or explore alternative sedimentation technologies such as tube settlers or inclined plate settlers to remain within limits.

Optimization of Coagulation Process. The identified optimum dosage of 2 mg/L of alum should be maintained to achieve effective turbidity removal while minimizing chemical use and environmental impact.

Environmental Considerations. While the carbon footprint and hardness induced by the coagulation process are relatively moderate, it is advisable to investigate eco-friendly coagulants or hybrid treatment methods that could further reduce environmental impacts.

Regular Monitoring and Adjustment. Continuous monitoring of raw water quality and treatment performance is essential to ensure that the designed parameters remain effective over time, especially in response to seasonal variations.

Public Awareness and Demand Management. With such high-water demand, awareness campaigns should be initiated to promote water conservation practices among the population to reduce stress on the treatment system.